LESSON PLAN

AIRCRAFT HUSBANDRY, CORROSION PREVENTION AND SURFACE PROTECTION (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Nuts, miscellaneous nuts, washers, Safety methods for turn buckles and various types of turn buckles.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Nuts, miscellaneous nuts, washers, Safety methods for turn buckles and various types of turn buckles.
5. Specific Objectives	25.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of standard and miscellaneous nuts/washers; Demonstration of turn buckle safety methods and assembly; Inspection of turn buckle types.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a nut in aircraft assembly?
(b) Name two types of washers used to distribute load.
(c) Why is the safety method for turn buckles critical in cable tensioning?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Standard nuts and miscellaneous nuts • Identification of bolts and nuts • Washer types and applications • Turn buckle safety methods • Various types of turn buckles	L & Ds	PPT, IB	15 min
(b)	• Detailed examination of turn buckle components • Safety inspection procedures for turn buckles • Correct installation techniques for turn buckles • Maintenance requirements for turn buckles • Troubleshooting common turn buckle failures	L & Ds	PPT, PDF	20 min
(c)	• Review of fastener standards and specifications • Comparison of turn buckle types and uses • Summary of safety protocols for turn buckles • Q&A session on identification and usage • Preparation for next lesson on cable identification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the safety method required before tightening a turn buckle.
(b) Differentiate between a standard nut and a miscellaneous nut based on usage.
(c) List three types of turn buckles and their specific applications.

12. Home Assignment and Follow Up	Prepare a brief note on the identification and replacement of swaged type terminals as covered in the next lesson plan (LP 04).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Cable identification, specifications, replacement, swaged type terminals splicing, swaged terminals splicing, swaged ball terminals, NICO press process and thimble eye.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min